AIDAN O'NEIL

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Houston, TX

EDUCATION

Rose-Hulman Institute of Technology **August 2022 – February 2026**
Bachelor of Science in Mathematics and Data Science and Artificial Intelligence, Minor in Economics *Terre Haute, Indiana*

PROFESSIONAL EXPERIENCE

Impiricus **Nov. 2025 – Present**
Data Engineer *Houston, TX*

- Lead engineer and architect of an interconnected analytics vertical (data warehouse, behavioral-scoring, and optimization-based scheduling), built and operated end-to-end across 5 production PostgreSQL/Aurora schemas and supporting 15+ engineers across 3 product teams.

HAIRGEL: Cross-Channel Dimensional Data Warehouse *Apr. 2026 – Present*

- Architected the system of record for cross-channel campaign engagement analytics: a 5-schema PostgreSQL/Aurora warehouse on a Kimball star schema with SCD Type-2 dimensions, a $\sim$12M-row partitioned fact table, and a $\sim$9M-row HCP dimension.
- Accelerated consumer queries 14$\times$–130$\times$ over the legacy OLTP path at exact 1:1 row parity, through a zero-failure hourly `pg_cron` ETL pipeline.
- Root-caused a multi-day silent production failure (PostgreSQL drops inherited group-role privileges after an in-procedure `COMMIT` while `has_table_privilege` still reports TRUE); resolved with direct grants and `SECURITY DEFINER` least-privilege execution.
- Designed a standardized 4-tier role-based access control (RBAC) framework across $\sim$10 services for 50+ users (SOC2-aligned governance); authored a unified all-channel event-log API and a daily automated agency click-export feed.

ION: Scheduling & Optimization Engine *Jan. 2026 – Present*

- Lead engineer; built and own end-to-end a 7-service, production-deployed scheduling and inventory-optimization platform ($\sim$28K LOC Python/Go, $\sim$700 tests, 18-table schema, OpenTofu/Terraform CI/CD).
- Replaced a manual allocation workflow with a two-pass economic optimizer — a HiGHS mixed-integer program with an LP-relaxation shadow-pricing feedback loop — scaling to 8M targets across 1000 campaigns over a 120-day horizon.
- Proved the program's integrality gap is exactly zero (min-cost network flow / total unimodularity), establishing the LP duals as exact competitive-equilibrium (Walrasian) prices; verified across 10,000+ instances.
- Named inventor on the ION scheduling-architecture patent application (filed 2026, patent pending); authored counsel review memos seeding a shadow-pricing continuation.
- Built the React/TypeScript Forecaster SPA (ALB–Cognito Google OAuth + RBAC) and own the production rollback/safety gate.

Cartograph: HCP Behavioral-Representation & Engagement Engine *May 2026 – Present*

- Designed and shipped a consumer-agnostic embedding engine: $d = 32$ learned provider representations plus an engagement scalar for 334K providers (Postgres/pgvector + HNSW), retrieval separation +0.547, treating cold-start ($\sim$95% of the network) as a first-class regime.
- Built a deterministic 4-signal inbound-reply scorer (two fine-tuned RoBERTa checkpoints over a 3-stage dispatch) replacing a 4-model GPT-4o chain, trained on an 11,731-row human-labeled corpus and deployed as a live FastAPI service.
- Measured engagement sparsity and burstiness across the provider network (median 1 substantive event/provider/year), redirecting the architecture to a count-based window; wired an integrality-safe engagement term into ION's optimizer.

Platform, Cross-Functional Leadership & IP *2026*

- Conceived and presented the firm-wide "Delivery Intelligence Platform" thesis (HAIRGEL $\rightarrow$ Cartograph $\rightarrow$ ION) at the company all-hands with a live ION demo; ran a cross-team ION code review for engineering, product, and QA.
- Designated cross-functional owner of the company-wide event taxonomy and engineering partner-of-record for Product on ION delivery; onboarded a new engineer onto ION code reviews and taught safe query/optimization patterns across teams.
- Built Peer Connect, a two-sided physician-to-physician matching product, on the shared embedding and optimization stack (winner of the 2026 Impiricus development hackathon; see Awards).

Booz Allen Hamilton **Jun. 2025 – Aug. 2025**
AI Engineering Intern *Fort Meade, Maryland*

- Co-architected (team of 4) a full-stack AI deployment framework for on-prem GPU clusters, enabling multi-model inference workflows without predefined specifications.
- Built a scalable RAG-enabled code retrieval system using GraphCodeBERT embeddings, designed to index repositories up to 1GB for contextual code generation and QA.

- Implemented Kubernetes-based deployment pipelines supporting concurrent LLM inference across 9 consumer GPUs (RTX 2080 $\times$4, RTX 2060 $\times$5), deploying up to 6 models simultaneously via vLLM.
- Developed 5 reusable LangGraph-based orchestration workflows, establishing modular components for future AI project teams.

O'Neil and Yuan Advisory **Apr. 2023 – Dec. 2024**

Co-Founder and Technical Lead *Houston, TX*

- Scoped, architected, and delivered 15 data pipelines for 11 nonprofit clients, handling datasets up to 40M historical records.
- Designed cloud-native data infrastructure, reducing client reporting cycles by 70% across manufacturing and retail engagements.
- Developed custom ETL systems and ML-based forecasting tools, managing end-to-end technical delivery and client-facing project sign-off.

AWARDS & RECOGNITION

- **"Outstanding Contributions to Impiricus" Award**, Data Engineering / Development team recognition (2026).
- **Winner, Impiricus Development Hackathon** (team award), 2026, for Peer Connect.

PROJECTS

Dynamic Network Modeling of Market Phase Transitions **Aug. 2025 – Feb. 2026**

- Developed a density-constrained dynamic graph model of financial markets using $N = 52$ liquidity-weighted S&P 500 constituents and sliding-window ($\omega = 60$) correlation tensors.
- Imposed fixed topological density ($\rho = 0.10$, $K = 132$ edges) to isolate structural changes independent of network sparsity, preserving signed edge information (correlation/anti-correlation).
- Defined and quantified an "Inhibition Vacuum" metric $I(t)$ capturing the collapse of anti-correlated edges preceding systemic hypersynchrony.
- Validated phase-transition significance using a 10,000-iteration stationary block bootstrap preserving temporal autocorrelation, observing statistically significant inhibition collapse ($p < 10^{-4}$) prior to the 2018 crash.
- Performed density sensitivity analysis, identifying a sparsity-dependent signal regime ("Sparsity Cliff" at $\rho \approx 0.12$) demonstrating structural signal degradation beyond biologically plausible connectivity thresholds.

Blockchain Fraud Detection via Graph-Based Feature Engineering **Feb. 2024 – Aug. 2024**

- Engineered spectral and structural graph features from a Bitcoin transaction network (203K nodes, 234K edges) for supervised fraud classification.
- Extracted centrality, community structure, and embedding-based features to capture latent transaction topology.
- Trained XGBoost and LightGBM classifiers achieving 82% accuracy, improving baseline performance by 8%.

TECHNICAL SKILLS

Languages: Python, SQL, Rust, Go, R, JavaScript/TypeScript, Bash/Zsh, LaTeX
Data Engineering & Infrastructure: PostgreSQL/Aurora, pg_cron, MongoDB, Redis, Apache Spark, Kubernetes, Docker, AWS (S3, Lambda, SageMaker, Glue, Athena, ECS, Cognito), OpenTofu/Terraform, pgvector, FastAPI, Serverless Architectures
Machine Learning & AI: PyTorch, scikit-learn, XGBoost, LightGBM, RoBERTa/Transformers, LangGraph, vLLM, RAG, GraphCodeBERT, Embeddings/HNSW
Optimization & Mathematics: Mixed-Integer Programming (HiGHS, OR-Tools), LP Duality & Shadow Pricing, Network Flows, Mechanism Design, Spectral Graph Theory, Control Theory, Network Science, Bayesian Inference
Developer Tools & Visualization: Git, Neovim, Linux/Unix, Tableau, Plotly, Seaborn, React

ADDITIONAL SKILLS

Languages: Mandarin Chinese (HSK 3.5)
Professional Operations: Technical Documentation, Stakeholder Management, Agile/Scrum, Cross-functional Team Leadership
Leadership & Activities: Shakespeare Club President; Phi Gamma Delta Member